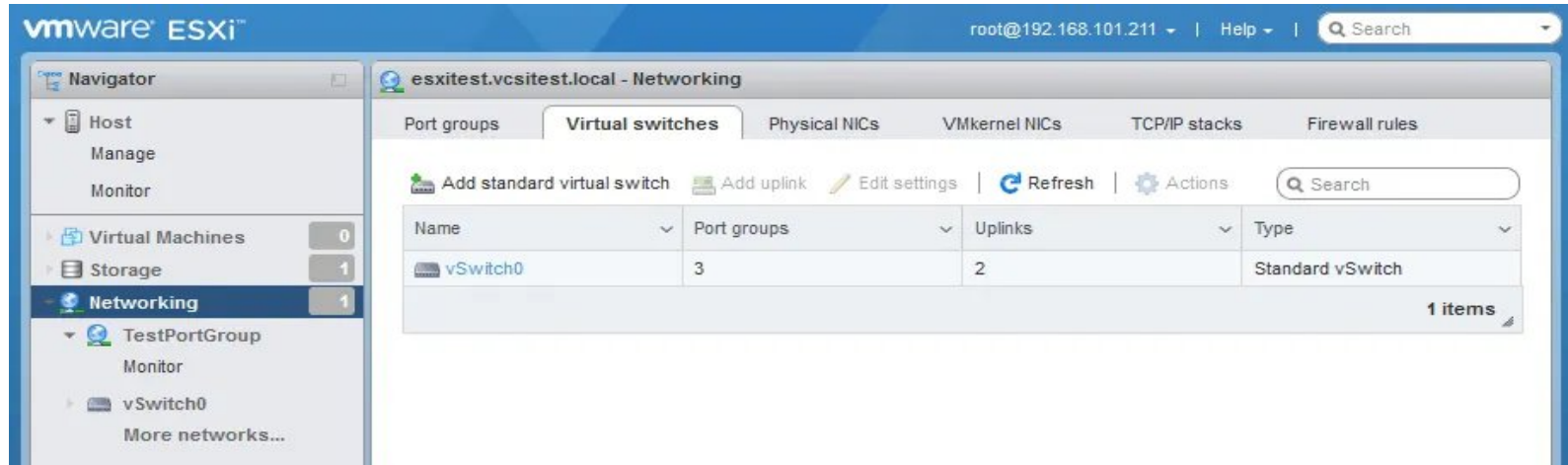


vSphere Network Concepts

Virtual Switches (vSwitches): is the key component enabling communication and data transfers between VMs in a VMware environment. A vSwitch is exactly what the name says: a virtual switch that is a software analog of a physical switch. Though not a hardware device, a virtual switch passes traffic using the same concepts used in the OSI networking model as physical switches. In other words, layer 2 frames (OSI data link layer) enter and exit the vSwitch the same way they would a physical switch.

One virtual switch (vSwitch0) is created by default when you install ESXi.



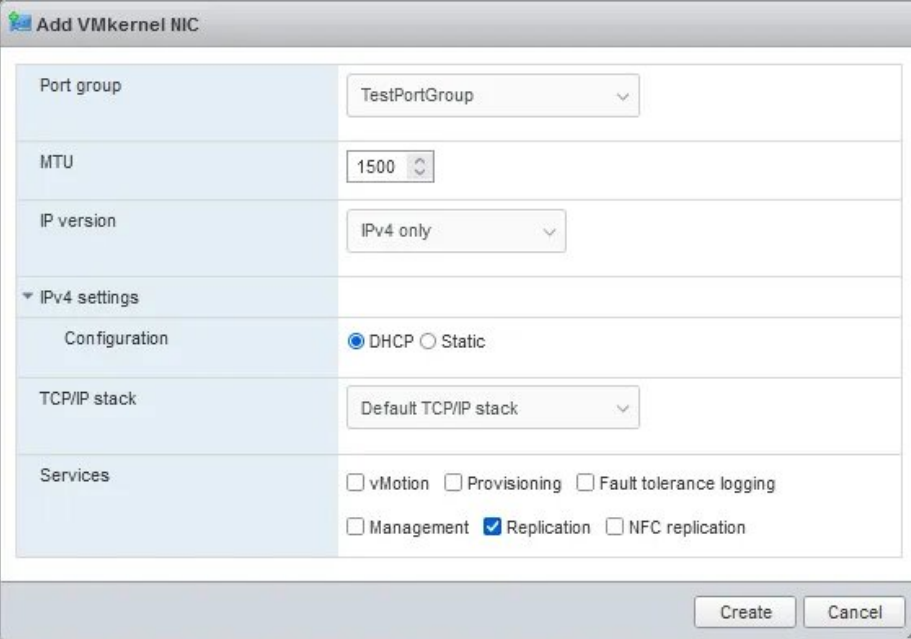
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vSwitch types: In vSphere, there are two types of virtual switches that can be provisioned, the vSphere Standard Switch (vSS) and Distributed Switch (vDS). The vDS requires Enterprise Plus Licensing to be available and is configured at the vCenter level. For the purposes of this post, we use vSS since they span the full range of licensed offerings from VMware and they don't require vCenter Server.

- **vSphere Standard Switch (vSS):** A basic, host-level virtual switch that provides network connectivity for virtual machines, managed individually on each ESXi host. It offers features like VLAN tagging, NIC teaming, and traffic shaping.
- **vSphere Distributed Switch (vDS):** A more advanced, centralized virtual switch managed via vCenter, spanning multiple ESXi hosts, offering advanced features like network I/O control, link aggregation, and centralized configuration across a cluster.

contd..

Virtual Ports: Virtual ports on the vSwitch act like the ports on a physical switch. Each VM that is configured with a virtual network adapter uses one of these virtual ports on the vSwitch. Also, there are **VMkernel ports**, which are a special kind of ports that talk directly to the ESXi host's VMkernel and are required to make storage, vMotion and management network traffic possible.



The screenshot shows the 'Add VMkernel NIC' configuration window. The settings are as follows:

Field	Value
Port group	TestPortGroup
MTU	1500
IP version	IPv4 only
IPv4 settings	
Configuration	☒ DHCP ☐ Static
TCP/IP stack	Default TCP/IP stack
Services	
☐ vMotion ☐ Provisioning ☐ Fault tolerance logging	
☐ Management ☒ Replication ☐ NFC replication	

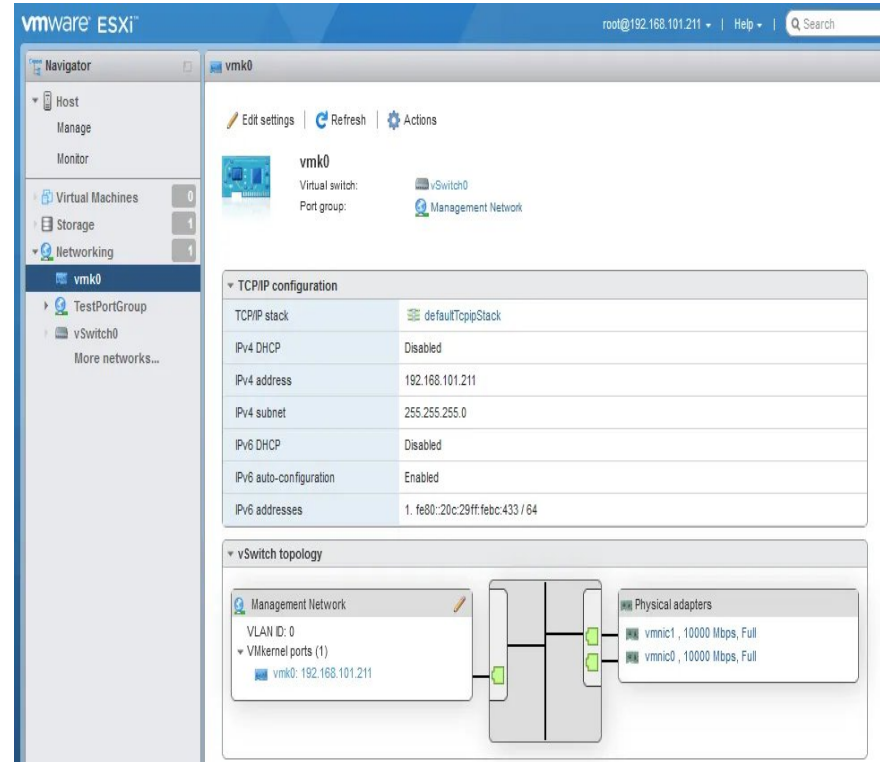
At the bottom right, there are 'Create' and 'Cancel' buttons.

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VMkernel: VMkernel is a specialized POSIX-like (Portable Operating System Interface) operating system in VMware ESXi that provides a layer of abstraction between the virtualized hardware and the physical hardware.

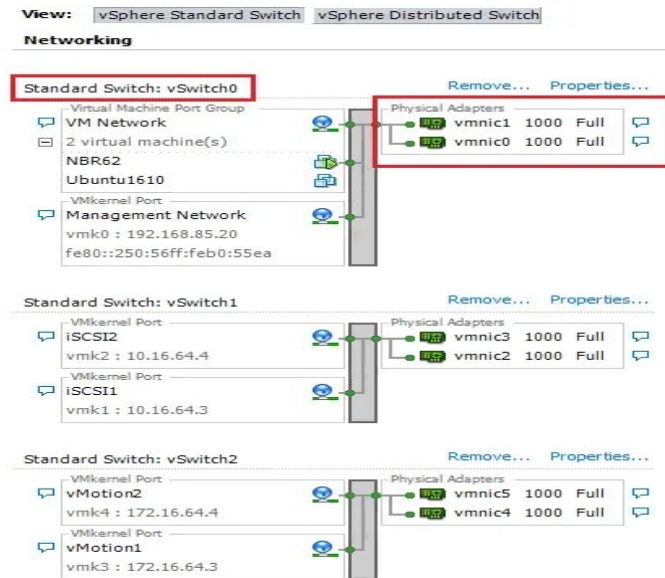
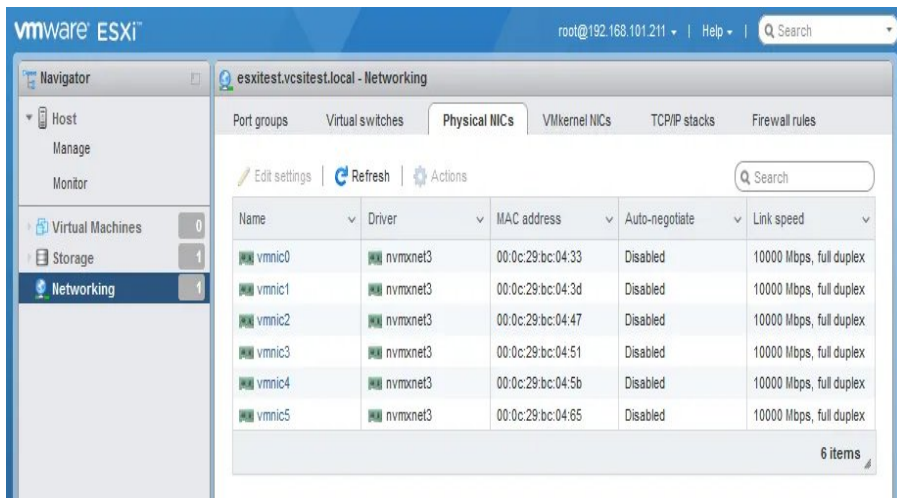
VMkernel is responsible for providing various services to virtual machines (VMs) running on the VMware ESXi hypervisor, including memory management, process scheduling, and I/O (input/output) management. It also provides network and storage connectivity to the VMs.

One of the key features of VMkernel is its ability to virtualize physical hardware resources, such as CPU, memory, and storage. This allows multiple VMs to share the same physical resources, enabling higher utilization rates and greater flexibility in resource allocation. VMkernel is required for DRS, which is a load balancing feature in vSphere.



contd..

Physical Uplinks: Virtual Switch traffic is uplinked on each ESXi host by the physical network cards in the ESXi host that are bound to that vSwitch in VMware. This provides the traffic path for traffic entering and exiting the vSwitch. The real physical interface on the ESXi host is mapped to a **vmnic** in ESXi. These mappings can be seen when managing the networking adapters on the ESXi host.



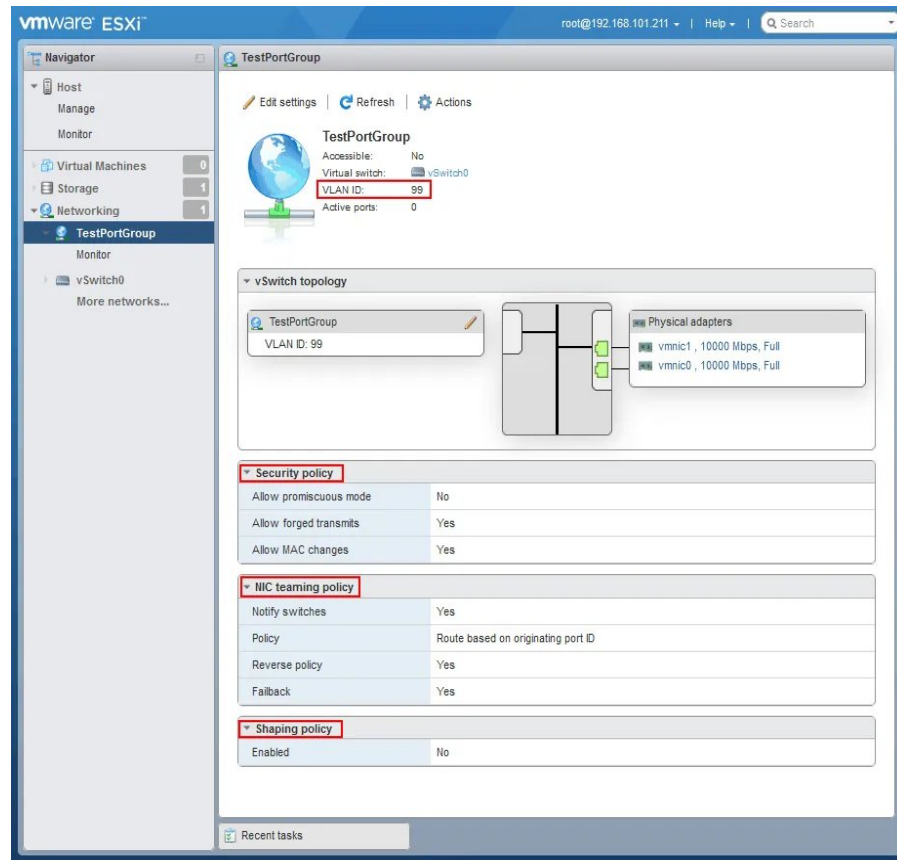
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VMware Port Groups: Port groups in VMware environments allow you to logically carve up virtual ports that are available on a particular vSwitch. We can apply traffic policy rules at the port group level: security rules and traffic shaping. Port groups are where you can also assign VLANs to your traffic.

A virtual switch can contain multiple port groups.

The two types of port group in VMware ESXi can be classified as follows:

- **Virtual machine port groups** – for connecting virtual machines to a virtual switch.
- **VMkernel port groups** – for connecting a VMkernel interface to a virtual switch.



contd..

VLANs: VLANs allow for segmenting networks into different pieces without having to physically separate networks. This is made possible by an additional **VLAN tag** being sent within the Ethernet frame header, which identifies which VLAN the traffic belongs to. This is often done for security and performance reasons. As the network becomes larger, broadcast traffic can grow exponentially. VLANs segment broadcast domains so that unwanted broadcast traffic is contained within a logical network.

Going back to our vSwitch, there are three ways we can set VLAN tags:

- **Virtual switch tagging** (VST) is setting VLAN tags at the port group level. This means setting the VLAN tag at the vSwitch level on the port group. This way the vSwitch is managing VLAN tags. This is perhaps the easiest and preferred way to introduce VLANs into a virtual environment.
- **External switch tagging** (EST) is setting VLAN tags on a physical switch. The physical switch is responsible for managing VLAN tags. In many VMware environments, if you have a dedicated storage switch, the ports will be set in **untagged** mode. This means that the physical switch is doing all of the tagging operations, and the virtual switch is not aware of this.
- **Virtual guest tagging** (VGT) allows the virtual guest VM to tag traffic and pass this along to the vSwitch, which then passes the VLAN-tagged frame up to the physical switch. This is the least common use case.

Network Services in vSphere

- **vSphere Network Services:**
 - **VMkernel Adapter:** A special type of network adapter used for vMotion, iSCSI, and other management traffic.
- **vMotion:** The process of live-migrating virtual machines between hosts without downtime.
- **Storage vMotion:** Moving VM disk files between datastores without interrupting the VM's operation.
- **Fault Tolerance:** Ensuring continuous VM availability by creating a secondary VM running in lockstep with the primary VM.

Hands-On Lab - Configuring Networking

- **Objective:** Configure virtual networks and network adapters.
- **Lab:**
 - **Activity 1:** Create a standard vSwitch and configure it with multiple NICs.
 - **Activity 2:** Create and configure a distributed switch and assign VLANs.
 - **Activity 3:** Set up VMkernel adapters for vMotion, iSCSI, and management traffic.